

Anmol Singh

Patiala, Punjab — asingh64_be24@thapar.edu — +91 9815739939

GitHub — LinkedIn — LeetCode — Portfolio

Professional Summary

Computer Science Engineer focused on full-stack development, AI/ML, deep learning, and computer vision, with hands-on experience building and deploying end-to-end applications. Proficient in the MERN stack, FastAPI, PyTorch, TensorFlow, CNNs, and modern transformer architectures, with experience integrating machine learning models into practical full-stack solutions. Strong problem-solving skills demonstrated through 200+ LeetCode problems covering Dynamic Programming, Divide and Conquer, Hash Tables, Trees, Arrays, Strings, Stacks, Linked Lists, and more.

Technical Skills

Languages: C/C++, Python, JavaScript, SQL, HTML, CSS

Machine Learning: PyTorch, TensorFlow, timm

Backend: FastAPI, Express.js, REST APIs

Frontend: React.js, Redux, RTK Query, Tailwind CSS

Databases & Cloud: MySQL, MongoDB, MongoDB Atlas, Cloudinary

Tools & DevOps: Git, GitHub, Postman, CI/CD

Projects

MedScan AI – Full-Stack Medical Imaging Diagnosis Platform — GitHub

- Built a full-stack medical imaging platform integrating four CNN-based diagnostic models to analyze X-ray, MRI, and CT scans for tuberculosis, brain tumors, lung cancer, and bone fractures.
- Developed an end-to-end workflow with React.js, Node.js, FastAPI, MongoDB, and JWT authentication, enabling scan uploads, model selection, prediction results, confidence scores, and downloadable reports.
- Achieved up to 100% accuracy across the diagnostic models, with the remaining models achieving 99.85%–96.87% accuracy, while providing per-class prediction probabilities for model outputs.

PlantDx – Fruit Plant Leaf Disease Classification Platform — GitHub

- Developed a full-stack plant disease classification platform integrating four transformer-based models to classify leaf diseases across apple, mango, cherry, and grape plants.
- Built an end-to-end prediction workflow using React.js, Node.js, FastAPI, MongoDB, and JWT authentication, supporting image uploads, model selection, multi-class predictions, confidence scores, and report generation.
- Achieved up to 100% accuracy across the transformer models, with the remaining models achieving 99.46%–98.75% accuracy for healthy and diseased leaf classification.

Education

B.E. in Computer Engineering

2024 – 2027

Thapar Institute of Engineering and Technology, Patiala

CGPA: 7.69